Toward Adaptive Check-In Workflows for Conversational Mental-Health Companions: A Scoping Literature Review and Research Agenda

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Abstract

Paper type. Scoping literature review and research agenda. This is preliminary research: the synthesis identifies a gap and proposes a candidate architecture and study design to fill it; no implementation has been undertaken and no human-subjects data are reported.

Conversational AI for mental health has matured rapidly. Peer-reviewed randomised trials show that cognitive-behavioural chatbots produce measurable symptom reductions, generative-AI chatbots produce large-effect improvements on clinical outcomes (Cohen's $d = 0.79$–0.90), and app-supported mental-health interventions as a class produce small-to-moderate effects ($g \approx 0.28$–0.58). What remains unsolved is the harder gap these tools do not close: no peer-reviewed deployment combines a *learned, longitudinal model of the client*, a *theory-grounded scheduling policy* that selects when to reach out and what to offer, a *multi-framework intervention library* the subject-matter expert authors and the policy selects from, and a crisis-safe conversational substrate — all evaluated on listener-centred outcomes. We synthesise 55 verified-DOI sources following PRISMA-ScR reporting practice as a scoping review, and organise the field around three pillars (*Sense*, *Decide*, *Act*) plus six cross-cutting concerns (therapeutic frameworks; wellbeing techniques; crisis detection; engagement; ethics; evaluation). The resulting synthesis sits at the crossroads of digital-mental-health biostatistics (JITAI and micro-randomized trials), HCI (conversational agents and the digital therapeutic alliance), clinical psychology (CBT, ACT, MI), and AI safety. The paper contributes three reusable artefacts: a human-centred taxonomy of the design space (Table 1), a comparison shortlist of defensible baselines for a future prototype (Table 3), and a subsystem-by-subsystem feasibility assessment (Table 4) showing that the integration gap can be addressed today using existing open models, methods, and evaluation instruments — without training a new foundation model. The review motivates a *Decide-first* research agenda: a contextual-bandit scheduler over an existing chat substrate, prospectively evaluated at $N \approx 30$–50 against a fixed-cadence baseline via a micro-randomized trial. This paper delivers the literature review and the agenda; the prototype and the trial are explicitly positioned as future work.

1 Introduction

Conversational AI for mental health has made rapid empirical progress. Fitzpatrick et al.'s Woebot RCT [1], Inkster et al.'s real-world Wysa evaluation [2], and the Heinz et al. NEJM AI Therabot RCT [3] collectively demonstrate that chat-substrate interventions can produce clinically meaningful symptom reductions and therapist-comparable alliance ratings. The meta-analytic baseline across 66 smartphone-intervention RCTs in the Linardon et al. synthesis [4] is $g \approx 0.28$–0.58 across depression, anxiety, stress, quality-of-life, and social-anxiety outcomes, with CBT-based apps with engagement support producing the largest effects. This empirical base is sufficient to justify continued deployment of supportive conversational companions; it is *not* sufficient to claim the field has found the right shape for them.

Three deeper gaps persist across the published corpus. (a) *No longitudinal learned model of the client.* Most chatbot deployments rely on static intake forms or implicit context windows; they do not learn what

this user is like over weeks or months, in ways that can transfer across sessions. (b) *No theory-grounded scheduling policy.* Delivery schedules are typically fixed (daily check-in, weekly session) or entirely user-initiated; the extensive JITAI / micro-randomized-trial / contextual-bandit / RL literature developed in mHealth biostatistics [5]–[7] has been applied brilliantly to physical activity and, in one recent study [8], to depression-relevant populations via text messaging, but has not been integrated into a multi-framework conversational mental-health companion. (c) *No multi-framework intervention library bound to a scheduling policy.* Most chatbots commit to a single therapeutic framework at design time; the SME who knows the audience has no seam through which to swap in ACT, MI, DBT skills, or CFT, and no scheduling intelligence that learns which move to select for which state.

This paper synthesises the literatures that bear on those three gaps and asks whether they can be filled with currently available methods. We organise the field around three pillars and six cross-cutting concerns:

- *Sense* (Section 3): how can a system build a longitudinal, psychologically-meaningful model of a client from the conversation itself (and, optionally, light-touch EMA)?
- *Decide* (Section 4): when and what should the system deliver — how does the field connect JITAI/MRT/CMAB policy mathematics to conversational mental-health content?
- *Act* (Section 5): how are chat-delivered interventions performing empirically, and what does the digital-therapeutic-alliance literature say about the mediator between delivery and outcomes?
- Cross-cutting: therapeutic frameworks (Section 6); wellbeing techniques (Section 7); crisis detection and safety (Section 8); engagement and attrition (Section 9); ethics, regulation, equity, and privacy (Section 10); and evaluation methodology (Section 11).

A future implementation need not be limited to language alone. In addition to chat and optional EMA prompts, watches and rings can provide an explicitly consented, lightweight sensing branch into the *Sense* layer, contributing signals such as sleep, activity, resting heart rate, heart-rate variability, temperature, and recovery proxies [9], [10]. Those signals do not establish a diagnosis on their own, but they can help detect deviations from routine and moments of heightened stress, fatigue, or dysregulation — exactly the kinds of state changes a *Decide* policy might use to time low-risk interventions such as distraction, grounding, breathing, meditation, or a reflective check-in.

Section 12 synthesises the integration gap through five operational criteria (C1–C5); Section 13 provides a subsystem-by-subsystem feasibility assessment; Section 14 states limitations; Section 15 offers a Decide-first prototype recommendation and introduces our candidate open-source implementation that occupies the gap the review identifies. Throughout the pillar sections we anchor the synthesis to concrete file-paths from that candidate implementation (`content/method/`, `content/techniques/`, `content/evidence/`, `content/safety/`, `src/lib/ai/`) so a reader can see exactly which literature justifies each design commitment; the system itself is named and described in Section 15.

Scope of the present paper. This is a *scoping literature review and research agenda*, not a system or evaluation paper. It is preliminary work: the synthesis identifies an integration gap and proposes a candidate architecture and an MRT-style study design to fill it, but no implementation has been undertaken, no users have been recruited, and no proximal or distal outcomes are reported. Where the paper anchors the synthesis to concrete file paths from a candidate open-source implementation (introduced in Section 15), those paths are illustrative of the architectural commitments the literature recommends; the prototype, the micro-randomized trial, and any clinical or behavioural outcomes are explicitly positioned as future work (Section 15).

Contribution. The novelty of this paper is not a new algorithm; it is the synthesis itself. Specifically, the paper contributes three reusable artefacts for the digital-mental-health community: (i) a human-centered taxonomy of the Sense→Decide→Act design space for conversational mental-health companions (Table 1, expanded through Section 3–11); (ii) a comparison shortlist of defensible baselines against which a future prototype can be benchmarked (Table 3); and (iii) a feasibility analysis showing which parts of a candidate integrated system are already well-supported by literature and which remain the key implementation risks (Table 4).

Research questions. This review answers eight research questions, summarised in Table 2.

Axis	Design positions covered by this review
Client model (Sense)	Static intake (most chatbots); EMA-augmented; passive-sensing-augmented; wearable-augmented (watch/ring); LLM-state-inference from chat. Scoped here primarily to the last two.
Input modality	Chat-only (baseline), chat+EMA, chat+passive-sensing, chat+wearable (e.g., watch/ring signals such as sleep, activity, HRV, temperature). Our candidate implementation is currently chat-only.
Policy class (Decide)	Fixed cadence; rule-based decision tree; contextual bandit; RL. We adopt CMAB / RL as the target policy class with MRT evaluation.
Action menu (Decide+Act)	Single-framework catalogue (Woebot, Wysa); multi-framework SME-authored catalogue (our candidate implementation); BCT-taxonomy-aligned.
Delivery substrate (Act)	Rule-based chat, fine-tuned Gen-AI chat, general LLM + prompt-engineered content (our candidate implementation uses a general LLM routed through an AI gateway).
Crisis layer	Keyword-only; keyword + moderation API; keyword + LLM safety classifier; C-SSRS-validated. Our candidate implementation ships keyword-only as Phase 0.
Therapeutic framework	CBT; ACT; MI; DBT skills; Person-Centered; CFT; SFBT. SME-selectable in `content/method/01-approach.md`.
Engagement strategy	None; reminders; gamification; adaptive schedule (the claim of this review).
Evaluation surface	Symptom scales (PHQ-9, GAD-7); WAI-SR digital therapeutic alliance; MRT proximal effects; retention curves.
Governance surface	Consent and disclosure; content-author separation; no-persistence default; open-source architecture.

Table 1: Design space for conversational mental-health companions synthesised by this review.

ID	Research question
RQ1	How can a longitudinal psychological model of the client be built from conversational data alone?
RQ2	What JITAI / contextual-bandit / RL methodology is available, and what has it already produced in mental-health-adjacent trials?
RQ3	What proximal and distal effect sizes should a chat-based mental-health intervention plan to detect?
RQ4	What digital-therapeutic-alliance instrument(s) should a prototype-scale evaluation use?
RQ5	Which therapeutic frameworks have sufficient meta-analytic support for the SME to select as a primary method?
RQ6	What is the evidence base for each of the four shipped wellbeing techniques in `content/techniques/`?
RQ7	How should crisis detection be layered on top of, or alongside, an LLM chat substrate?
RQ8	What engagement / adherence baseline should an adaptive scheduling policy be benchmarked against?

Table 2: Research questions guiding this review.

2 Methodology

This is a *scoping review* that adopts the Arksey–O'Malley five-stage framework [11] with the methodological refinements of Levac et al. [12], and follows the PRISMA extension for scoping reviews (PRISMA-ScR) [13] for reporting. We additionally align with PRISMA 2020 [14] where applicable. It is not a fully systematic review in the clinical-trials sense; the heterogeneity of evaluation designs across HCI, mHealth biostatistics, clinical psychology, and natural-language processing does not support a single quantitative pooling.

Citation discovery was performed through a verified-DOI literature-search service (Scite, `https://api.scite.ai`) that returns DOI-verified metadata, full-text excerpts where available, smart-citation context (supporting / contrasting / mentioning / unclassified), and editorial-notice flags (retractions, concerns, errata, corrections). Industry-press sources were retrieved separately and treated as dated snapshots. Searches were performed in April 2026 with no end-date filter and a soft start date of 2014 (the year of the first formal JITAI position paper) unless an earlier foundational reference was needed (e.g. Arksey and O'Malley's 2005 framework [11], Rogers' core-conditions paper, or Beck's foundational CBT manuals).

Boolean search queries. Searches were structured around twelve Boolean queries spanning the three pillars and six cross-cutting concerns of the synthesis:

- Q1 (*Decide*): `"just-in-time adaptive intervention" AND mental`
- Q2 (*Decide*): `"micro-randomized trial" AND (mhealth OR "digital health")`
- Q3 (*Decide*): `"contextual bandit" AND ("digital health" OR mhealth OR "behaviour change")`
- Q4 (*Act*): `("chatbot" OR "conversational agent") AND ("depression" OR "anxiety" OR "mental health") AND (RCT OR trial)`
- Q5 (*Act*): `"large language model" AND ("mental health" OR "psychotherapy" OR "counselling")`
- Q6 (*Act*): `"therapeutic alliance" AND (chatbot OR "conversational agent" OR AI)`
- Q7 (*Sense*): `"ecological momentary assessment" AND (depression OR anxiety OR affect)`
- Q8 (*Frameworks*): `"behaviour change wheel" OR "behaviour change technique" AND digital`
- Q9 (*Frameworks*): `("acceptance and commitment therapy" OR "motivational interviewing" OR "cognitive behavioural therapy") AND (digital OR app OR chatbot)`
- Q10 (*Techniques*): `"slow breathing" AND ("heart rate variability" OR vagal)`
- Q11 (*Techniques*): `"self-compassion" AND (intervention OR trial)`
- Q12 (*Safety*): `"Columbia Protocol" OR "Columbia Suicide Severity Rating Scale" OR "Zero Suicide"`

Each query returned an ordered candidate list under default Scite relevance ranking; queries overlapped, and the candidate set was iteratively pruned during screening rather than fixed ex ante, so we report only the audited final set rather than a strict record-flow count, in line with the Arksey–O'Malley fifth stage [11], [12].

Inclusion criteria. Verified DOI; peer-reviewed venue (or recent foundational arXiv preprint with ≥ 5 citing publications); topical relevance to one of the research questions in Section 1; sufficient methodology detail to support the claim being cited.

Exclusion criteria. Retracted publications or papers flagged with editorial concerns (checked via Scite `retraction_notices` and `has_retraction` / `has_concern` filters); non-English publications without published translation; duplicates of arXiv preprints already cited via their archival venue. Title/abstract screening removed work outside scope, and full-text screening (where available via the Scite `fulltextExcerpts` field, or via the publisher access link otherwise) retained the records cited in this paper.

The final synthesis draws on the verified-DOI sources listed in the references. Sources were synthesised narratively; the LaTeX source and bibliography are released so the synthesis is reproducible from a clean TeX Live install (`cd paper/survey && make`).

3 Pillar A: Sense — Learning the Client Over Time

The Sense pillar asks: *how can a system build a reliable, longitudinal, psychologically-meaningful model of an individual client without crossing into surveillance, and using only the channels actually available to a conversational mental-health companion*? For a chat-based companion this means, primarily, *the conversation itself*, and (optionally, in future deployments) light-touch ecological momentary assessment (EMA) prompts and, where a user explicitly opts in, lightweight wearable-derived state signals from devices such as watches and rings. Heavy passive sensing is held out of scope by design: SME-authored boundaries in a companion's content layer typically forbid persistent capture without consent (see `content/safety/boundaries.md` in our candidate implementation), and the legal and sociological cost of that data class is high (Section 10). The literature reviewed here therefore divides into four threads, ordered from most to least directly applicable to a chat-only baseline.

Ecological momentary assessment as the conceptual anchor. EMA was formalised by Stone and Shiffman in 1994 and matured into a mainstream clinical-assessment method through the 2000s; the most useful synthesis for a digital-mental-health implementer is [15], the special-section introduction in *Psychological Assessment* that defines EMA as in-the-moment, real-world, repeated sampling of states and behaviours.

EMA's two empirical commitments—ecological validity and the avoidance of retrospective recall bias—transfer directly to a chat-based check-in workflow. [16] extend the EMA paradigm into *ecological momentary interventions* (EMI) for depression and anxiety: brief, in-context interventions delivered in response to a detected state. EMI is the conceptual ancestor of the JITAI literature in Pillar B; together they motivate the Sense→Decide handoff this synthesis foregrounds.

Passive sensing as the upper bound. Two reviews delineate the modern landscape of passive sensing for mental-health monitoring. [17] systematise the smartphone-only literature and find that step counts, location clusters, screen time, and inter-event timing yield modest but reproducible signal on depressive symptoms; reliability is bounded by sensor noise, user-context drift, and missing-data patterns that correlate with the clinical state being measured. [9] broaden the picture to wearable and environmental sensors and confirm that multimodal passive sensing outperforms any single modality but at substantially higher infrastructural and ethical cost. [18] provides the influential vision paper for "digital phenotyping"—the proposition that smartphone-derived data streams (sensors, voice, human–computer interaction) can serve as a global tool for psychiatry. Empirically, the Dartmouth StudentLife project [19] is the canonical demonstration: sensor-derived behaviour explains a non-trivial share of variance in self-reported stress, sleep, mood, and sociability across a 10-week academic term, with a publicly released dataset that has anchored hundreds of follow-on studies. The same group's COVID-cohort extension [20] provides the strongest within-person evidence to date that passive sensing + EMA detects state change in real time. We treat these as the upper bound on what Sense *could* contribute if a chat-only companion eventually adds an EMA layer; they are not a baseline a chat-only architecture can match. For the specific agenda of this paper, watches and rings are especially interesting because they can contribute continuous signals related to sleep, activity, resting heart rate, heart-rate variability, temperature, and recovery proxies [9], [10]. Those signals do not yield a diagnosis on their own, but they can provide useful evidence of heightened stress, reduced recovery, sleep disruption, or deviations from routine — exactly the kinds of state changes a Decide layer could use to time low-risk interventions such as distraction, grounding, paced breathing, or meditation prompts.

Natural-language signal from conversation. Closest to the chat substrate of a conversational companion is the literature on inferring mental-health states from text. The most thoroughly studied corpus is Reddit. [21] provides a scoping review of depression and anxiety studies using Reddit as a data source; the field has converged on transformer-based classifiers that distinguish sub-clinical and clinical signal at population scale, while remaining poorly calibrated at the individual level and vulnerable to selection bias (only Reddit users; only those who self-disclose). Closed-vocabulary lexical methods such as Linguistic Inquiry and Word Count (LIWC) remain useful where interpretability matters. As an illustrative recent example, [22] run a case-control LIWC analysis of teenagers' Instagram captions in the three months before suicide vs. matched living controls and find significant differences in word-per-sentence length and references to sadness, drives, leisure, and several function-word categories. The clinical implication for a chat-based companion is bounded but real: a small, transparent set of lexical features can flag salient state shifts without invoking an opaque classifier. The Sense layer in the integrated system this review motivates should therefore favour interpretable lexical features over deep classifiers wherever both options exist, with deep classifiers reserved for cases where lexical methods underperform.

LLM-based state inference. The newest thread is the use of large language models themselves as state-inference engines—the only thread that does not require a chat-based companion to ingest channels outside its existing chat data flow. The *Mental-LLM* evaluation of [23] is the most comprehensive published comparison: across multiple mental-health prediction tasks (depression detection, stress detection, suicidal ideation), instruction-tuned smaller models (Mental-Alpaca, Mental-FLAN-T5) outperform the best zero-shot prompt of GPT-3.5 ($\sim$10.9% balanced-accuracy gain), and approach the performance of larger models such as GPT-4. The position paper of [24] surveys the broader opportunity-and-risk surface for LLMs in mental health: meaningful capacity for both listening and basic state inference, but persistent risks around hallucinated specificity, sycophancy, and the absence of clinical escalation paths. For an LLM-routed chat architecture (the pattern used by our candidate implementation in `src/lib/ai/`) this is particularly germane: a chat substrate that already routes through a general-purpose LLM is, in principle, also a sense substrate at no additional cost; the architectural commitment is to call out that secondary use explicitly, log it, and bound it.

Critical caveats. We close the pillar with a deliberately uncomfortable caveat. [25] provide a sociological critique of digital phenotyping that rejects the implicit framing of mental-health data as a passive "signal" to

be "mined". The Sense layer in any system built on this synthesis must contend with three of their points: (i) assumptions of normality embedded in the algorithm; (ii) the limits of digital data as a proxy for social life; (iii) the risk of biological or computational language reifying a state the user might frame differently. The implication for a Sense-equipped conversational companion is concrete: the policy described in Pillar B must be auditable and reversible by the user; the SME's framework should drive what the state representation *means* (see `content/method/` in our candidate implementation); and the integrated-system criteria C1–C5 of Section 12 explicitly require a learned client model that the user can inspect and reset.

4 Pillar B: Decide — When and What to Reach Out With

The Decide pillar is the centre of gravity for this review and for the gap-filling architecture our candidate implementation proposes (Section 15). The question is structurally identical across two decades of digital-health research: *given a longitudinal model of the client (Pillar A), and a library of available intervention components (Pillars C and beyond), what should be delivered and when*? The answer the field has converged on is a *just-in-time adaptive intervention* (JITAI): a sequence of decision rules whose inputs are the user's evolving state and whose outputs are intervention actions selected from a catalogue. The literature divides into four threads: the JITAI conceptual framework, the micro-randomized trial (MRT) as its evaluation design, contextual bandits and reinforcement learning (RL) as its policy class, and the behaviour-change taxonomies that name what the action menu actually contains.

JITAI: the conceptual framework. The canonical synthesis is [5]: a JITAI is operationalised by four constructs — *decision points* (when to ask), *tailoring variables* (what state to read), *decision rules* (the policy mapping state to action), and *intervention options* (the action menu, ranging from no action through low- and high-dose components). The motivation, in their language, is to deliver *the right type of support, in the right amount, at the right time, to the right person* — and, equally important, to provide nothing when the time is wrong, since notification fatigue and intervention burden are the dominant failure modes of digital-health deployments. [26] extend this picture in 2025 with a qualitative systematic review of JITAI deployments in mental health specifically and document a striking pattern: the JITAI literature is mature for physical-activity outcomes but remains thinly populated for clinical mental-health outcomes. This gap is structural, not accidental, and it is precisely the gap that motivates the integration framing of Section 12. In a wearable-augmented deployment, tailoring variables need not be limited to what the user just said in chat: they could also include recent deviations in sleep, activity, or HRV-derived recovery, so that decision points become sensitive to moments when the body appears stressed, dysregulated, or unusually depleted [9], [10].

Micro-randomized trials: the evaluation design. The evaluation methodology that lets a JITAI be developed empirically rather than guessed at is the micro-randomized trial (MRT) of [6]: each individual is randomized hundreds or thousands of times across the study, so the proximal causal effect of each push intervention component can be estimated under a clean identification assumption. The associated statistical machinery is mature: [27] provide the test statistic and sample-size calculator for proximal-effect tests; [28] define a time-varying standardized effect size and demonstrate that Cohen's small/medium/large heuristics are typically too generous for proximal mHealth outcomes, motivating realistic effect-size budgets; [29] contribute the practical implementer's guide to MRT data systems — feature selection, randomization agents, missing-data anticipation, and a reproducibility checklist. We adopt the MRT as the recommended evaluation design for the Decide-first prototype recommended in Section 15.

Contextual bandits and reinforcement learning as the policy class. The next layer is the policy itself: rule-based decision trees suffice for prototypes, but the research frontier has moved to contextual bandits and full RL. [7] provide the canonical demonstration: an online-learning RL algorithm embedded in HeartSteps V2 that decides, five times per day, whether to deliver a context-tailored physical-activity suggestion; the algorithm continuously improves its policy from the data the trial generates, and is the strongest published evidence that online RL can be deployed inside a real clinical trial without sacrificing the identifiability of proximal effects. The empirical anchor for this RL algorithm is the V1 efficacy trial of [30], which shows both that contextually-tailored suggestions produce a measurable proximal step-count effect, *and* that the effect attenuates over time — the most direct empirical motivation for *adaptive* (vs. fixed-cadence) scheduling. [31] extend the picture to mental-health-adjacent content: a doubly-robust offline contextual-bandit recommender for emotion-regulation strategy selection in $n = 114$ socially anxious participants. They

demonstrate that learning a policy over an intervention catalogue is feasible offline from naturalistic data — the most directly relevant published precedent for a catalogue-over-techniques policy in a conversational mental-health companion. Most recently, [8] report the DIAMANTE RCT (n=168, multilingual) comparing a Thompson-sampling contextual-bandit messaging arm against random and weekly-control arms over six months in adults with diabetes *and* depression: the RL arm produced a 19% step-count gain vs. 1.6% (random) and 3.9% (control) ($p < .001$). DIAMANTE is the first large RCT to test an adaptive RL-driven schedule against a randomly-timed baseline in a population with mental-health-relevant comorbidity, and it is the closest existing benchmark for the kind of evaluation the Decide layer of an integrated conversational companion would warrant.

Behaviour-change taxonomies: naming the action menu. A learned policy is only as useful as its action vocabulary. The behaviour-change literature provides two complementary taxonomies that make this vocabulary explicit. The Behaviour Change Wheel of [32] places nine intervention functions (education, persuasion, incentivisation, training, enablement, coercion, restriction, environmental restructuring, modelling) and seven policy categories around the COM-B behaviour system (capability–opportunity–motivation → behaviour); the BCT Taxonomy v1 of [33] extends this to 93 hierarchically clustered behaviour-change techniques. For any SME-authored companion these taxonomies have a concrete mapping: in our candidate implementation, the four currently-shipped wellbeing techniques in `content/techniques/` (paced breathing, 5-4-3-2-1 grounding, cognitive reframing, self-compassion break) sit cleanly inside specific BCT clusters (4. Shaping knowledge; 11. Regulation; 13. Identity), and the SME's authoring conventions in `content/method/` sit inside the BCW intervention-function categories of *enablement* and *persuasion*. A Decide policy built on top is therefore not a black-box optimiser over an opaque catalogue; it is a policy over a *taxonomically labelled* catalogue, which is auditable by both the SME and (as discussed in Section 10) the user.

Pillar B summary. The literature delivers four assets that a Decide layer in a mental-health companion can compose without inventing new theory or methodology: a conceptual framework ([5]); an evaluation design and the statistics to power it ([6], [27]–[29]); a policy class with both online ([7]) and offline ([31]) precedent and at least one mental-health-relevant RCT ([8]); and a shared vocabulary for the action menu ([32], [33]). The remaining gap is the *combination*: no published deployment applies these four assets to a multi-framework, SME-authored, chat-delivered mental-health companion. That combination is the integration claim of Section 12.

5 Pillar C: Act — Conversational Delivery

The Act pillar concerns how an intervention selected by the Decide policy is actually delivered to the client. In our candidate implementation this is fixed by architecture: delivery is via an LLM-powered chat substrate (`src/lib/ai/`) configured with an SME-authored system prompt assembled from the `content/` Markdown (cf. `src/lib/ai/system-prompt.ts`). The literature divides into three threads: the RCT evidence for individual conversational agents; the meta-analytic landscape; and the therapeutic-alliance literature that mediates clinical effectiveness.

Conversational agents with peer-reviewed RCT evidence. [1] is the canonical published RCT of a CBT-based conversational agent for depression and anxiety (n=70, 2-week intervention): Woebot produced a significant PHQ-9 reduction over an information-only control, with engagement averaging 12 sessions per participant. The Woebot trial established three clinically relevant facts the field has continued to rely on: a rule-based chat substrate can deliver short-cycle CBT content with measurable symptom effects; process factors (empathy, pacing, timing of reflections) are at least as acceptable to users as content factors; and the main caveat is engagement attrition. The Wysa mixed-methods real-world evaluation of [2] extends the picture with a larger sample ($n \approx 129$ high-engagement) and shows that self-reported depression improvements scale sharply with engagement level — the coupling of outcomes to engagement that motivates the scheduling-intelligence argument of Pillar B.

The single strongest piece of recent evidence is the NEJM AI Therabot RCT of [3], published in March 2025. Therabot is an expert-fine-tuned generative-AI chatbot; in a preregistered n=210 RCT it produced large effect sizes (Cohen's d=0.79–0.90) on MDD, GAD, and clinically-high-risk feeding/eating disorder symptoms over a four-week intervention with an eight-week follow-up, with participants rating the therapeutic alliance as *comparable to human therapists*. Therabot is the first peer-reviewed RCT to show that a Gen-AI chatbot

can produce clinically meaningful effects on primary mental-health symptoms using chat as the only modality. It is also the clearest existing comparator for a prototype-scale deployment of an integrated companion, and we adopt it as the Act baseline in Section 12's comparison shortlist.

Meta-analytic landscape. The broader picture is summarised in the World Psychiatry meta-analysis of [4]: across 66 RCTs of app-supported smartphone interventions for mental-health problems, smartphone interventions significantly outperformed control conditions on depression (g=0.28), generalized anxiety (g=0.30), stress (g=0.35), quality of life (g=0.35), general distress (g=0.40), social anxiety (g=0.58), and positive affect (g=0.44), with effects robust to sensitivity analyses. Two sub-findings are particularly load-bearing for the present synthesis. First, CBT-based apps outperformed non-CBT-based apps — consistent with the SME's option in our candidate implementation's `content/method/01-approach.md` to pick CBT as the primary framework. Second, apps that offered *engagement reminders and professional guidance* produced the largest effects. This second finding is the empirical anchor for the Decide pillar itself: scheduling intelligence is not a cosmetic feature, it is on the critical path to clinical gains.

Therapeutic alliance with a conversational agent. The mediator between delivery and outcomes is the working alliance. [34] provide the first focused review of the therapeutic-alliance literature in digital mental-health apps and identify, as the central methodological gap, the lack of a purpose-built digital-alliance measure; simply replacing "therapist" with "app" on the Working Alliance Inventory is insufficient because several face-to-face alliance facets do not port. [35] introduce the JMIR Mental Health special issue on the digital therapeutic alliance (DTA) and argue for purpose-built measures. The most recent comprehensive synthesis is the 2025 integrative review of [36] (28 included studies), which proposes a working DTA definition built on four elements: goal alignment, task agreement, therapeutic bond, and user engagement — the same four elements the Decide policy of an integrated conversational companion operates over (what goal to pursue, what task/technique to offer, how warmly to speak, when to prompt). Empirically, [37] demonstrate that the chat interaction modality itself (click-based menu vs. free-text input) has measurable alliance consequences: writing-based interactions yield higher bonding, click-based give easier on-ramp. The design implication is that an integrated companion's chat substrate should support both modalities fluidly as the Decide policy learns what works for the individual client.

Commercial landscape and adjacent industry solutions. The related-solution space is broader than the peer-reviewed chatbot literature. Mainstream consumer products already cover several slices of the design space this review analyses, even if they do not combine them into a single integrated system. In the Apple ecosystem, the *Journal* app provides context-aware, multimodal reflective prompting from recent photos, workouts, music, locations, and other phone activity,[1] while Apple Watch's *Mindfulness* features deliver short *Reflect* and *Breathe* sessions with reminders and heart-rate feedback,[2] and the *State of Mind* feature supports momentary-emotion and daily-mood logging linked to sleep, exercise, and other lifestyle factors.[3] On the Android / Google side, Fitbit and Google Pixel Watch provide a stress-management stack that includes stress scores derived from sleep, activity, heart rate, HRV-related signals, mood logging, breathing exercises, and body-response alerts,[4][5] which is the closest mainstream product expression of the wearable-augmented Sense→Decide logic proposed in this review. Major commercial wellness apps such as *Headspace* and *Calm*, both available across iOS and Android, contribute the meditation / sleep / breathwork side of the landscape through large content libraries, daily programs, stress-relief tools, and in Headspace's case an AI companion for support and content recommendation.[6][7] We treat these products as *industry snapshots*, not as load-bearing scientific evidence, but it is important that the paper accounts for them explicitly: together they show that the market has already validated user interest in journaling, breathing, mindfulness, stress reflection, and wearable-triggered self-regulation. What remains absent, in both academy and industry, is the specific combination this review argues for: a psychologically-grounded, multi-framework companion that learns the client over time and uses that model to schedule and tailor check-ins adaptively.

Pillar C summary. The literature provides four usable assets for a conversational mental-health companion:

[1] `https://www.apple.com/newsroom/2023/12/apple-launches-journal-app-a-new-app-for-reflecting-on-everyday-moments/`
[2] `https://support.apple.com/guide/watch/practice-mindfulness-apd371dfe3d7/watchos`
[3] `https://support.apple.com/guide/watch/log-your-state-of-mind-apd7de0f5610/watchos`
[4] `https://support.google.com/fitbit/answer/14237928`
[5] `https://blog.google/products/fitbit/measure-stress-fitbit-google-pixel-watch`
[6] `https://www.headspace.com/how-it-works`
[7] `https://www.calm.com/`

(i) RCT-validated precedent for conversational-agent delivery of mental-health interventions ([1]–[3]); (ii) meta-analytic evidence that CBT-based apps with engagement support produce the largest effects, with effect sizes on the order of g=0.3–0.6 ([4]); (iii) an evolving digital-therapeutic-alliance literature that maps cleanly onto the goal/task/bond/engagement axes a Decide policy operates over ([34]–[36]); and (iv) interaction-modality design evidence ([37]). What the literature does *not* provide is a published system that combines any of these with a *learned scheduling policy* (Pillar B) over a *multi-framework SME-authored method catalogue* (Section 6–7). That combination is the integration claim of Section 12.

6 Cross-Cutting: Therapeutic Frameworks

An SME-authored conversational companion must commit to one or more therapeutic frameworks — CBT, ACT, MI, DBT skills, Person-Centered, CFT, SFBT, or a blend. In our candidate implementation this commitment lives in a single editable file (`content/method/01-approach.md`); other companions make analogous commitments at design time. The literature-review question here is narrow: which frameworks are evidence-based choices for a supportive (non-clinical) companion, and what effect-size budgets should be expected? We cite the three most authoritative meta-analyses; the remaining frameworks (DBT skills, Person-Centered, CFT, SFBT) each have their own meta-analytic anchors that can be added if the SME selects them, but we keep this section compact because it is not a differentiating contribution of this survey.

Cognitive Behavioural Therapy is the best-evidenced framework by a substantial margin. [38] review 106 meta-analyses spanning substance-use disorder, psychosis, mood disorders, anxiety, somatoform disorders, eating disorders, insomnia, personality disorders, anger, criminal behaviours, general stress, pain and fatigue, and pregnancy-related distress, and find that CBT has the strongest support for anxiety disorders, somatoform disorders, bulimia, anger-control, and general stress; in head-to-head comparisons CBT outperforms or matches alternatives in 7 of 11 reviews. The practical implication for a conversational companion is that CBT is a safe default framework choice, and it aligns cleanly with the Decide pillar's interest in structured, time-limited intervention components.

Acceptance and Commitment Therapy is supported by the [39] meta-analysis of 39 RCTs (n=1,821): ACT outperformed controls at posttreatment (g=0.57), waitlist (g=0.82), placebo (g=0.51), and treatment-as-usual (g=0.64), and showed no significant difference from CBT on primary outcomes. ACT is a defensible alternative or complement to CBT for an SME whose audience benefits from acceptance-and-values framing over cognitive-restructuring framing.

Motivational Interviewing has 25 years of accumulated evidence synthesised by the [40] meta-analysis of 119 studies across substance-use, health-behaviour, gambling, and treatment-engagement outcomes: durable small-effect gains (g=0.28) over weak comparisons and non-significant results against specific active treatments. The implication is narrower-but-clear: MI is a strong match where the task is engagement and ambivalence-resolution (both plausible primary tasks for a supportive companion) rather than acute symptom reduction.

Each of these frameworks maps cleanly onto clusters in the Behaviour Change Technique Taxonomy of [33] and intervention-function categories of the Behaviour Change Wheel [32] — a mapping that allows a Decide policy to index over an SME-authored catalogue.

7 Cross-Cutting: Wellbeing Techniques

Our candidate implementation currently ships four starter wellbeing techniques in `content/techniques/`: *box breathing*, *5-4-3-2-1 sensory grounding*, *cognitive reframing*, and the *self-compassion break*. Each is a candidate *action* the Decide policy can select. This section documents the evidence base for each so the SME's `content/evidence/references.md` has a literature anchor; an SME integrating a different catalogue would substitute the analogous evidence for their techniques.

Paced/slow breathing. The mechanistic case for paced breathing rests on vagal cardiac control, summarised in the modern psychophysiological-methodology reference of [10]: HRV-derived indices of cardiac vagal tone are associated with self-regulation at the cognitive, emotional, social, and health levels, and slow respiration at resonance frequency (approximately 6 breaths/min) amplifies respiratory sinus arrhythmia and vagally-mediated HRV. The clinical-outcome meta-analysis of HRV biofeedback by [41] pools effects across anxiety,

depression, PTSD, and performance outcomes and finds moderate clinically-relevant gains. Box breathing (4-4-4-4) is a specific variant; while it is not the resonance-frequency pattern of the Lehrer protocol, it shares the slow-breathing mechanism and is the technique shipped in `content/techniques/box-breathing.md` in our candidate implementation.

5-4-3-2-1 sensory grounding. Unlike the other three techniques in the starter set, 5-4-3-2-1 grounding does not have a single-RCT evidence anchor. It is a standard phase-1 stabilisation skill drawn from the broader sensory-grounding tradition in trauma-focused therapy, best contextualised by [42]'s discussion of imagery and grounding in phase-1 treatment of complex dissociative disorders. The clinical implication for an SME is important: the technique is used widely in clinical practice as a low-risk sensory anchoring exercise, but its evidence base is weaker than the other three, and the SME should treat it as a symptom-management tool rather than an outcome-producing intervention. The `Avoid for` section of `content/techniques/grounding-54321.md` should accordingly be inclusive.

Cognitive reframing. Cognitive restructuring is the core mechanism of CBT and its evidence base is therefore the CBT meta-analytic base of [38] cited in Section 6. The candidate implementation's `content/techniques/cognitive-reframing.md` operationalises this as a short, reflective exercise that surfaces the thought under a feeling and invites a softer, still-true alternative. The SME should be aware that while cognitive restructuring is well-evidenced, un-timed or premature reframing can be experienced as invalidating; the technique's own `Avoid for` section already flags this, and the Decide policy should treat "timing-of-reframe" as a learnable parameter.

Self-compassion break. The self-compassion break derives from the Mindful Self-Compassion (MSC) program developed by Neff and Germer. The pilot RCT of [43] shows significant gains in self-compassion, mindfulness, and wellbeing in an 8-week MSC program. The single-session self-compassion break shipped in `content/techniques/self-compassion-break.md` is a bite-sized version of the same construct and inherits a substantial downstream evidence base (a family of RCTs subsequently built on the MSC protocol and its components). Like cognitive reframing, the critical caveat is timing: the self-compassion construct is well tolerated on average but can trigger backdraft (defensive resistance) in users with strong self-critical schemas, which is why the Decide policy's "when to offer" decision matters.

8 Cross-Cutting: Crisis Detection and Safety

A Phase-0 safety stack for a chat-based mental-health companion can be built from two layers: (i) a keyword-matching crisis detector (e.g. `content/safety/crisis-keywords.md` in our candidate implementation) that raises a crisis banner in the chat UI when a substring match is detected; and (ii) system-prompt reminders (`content/safety/disclaimers.md`, `content/safety/boundaries.md`) that define hard "no" behaviours and require explicit AI-disclosure. The governing rule is that *false positives are acceptable; missed signals are not*. This section anchors such a stack against four literatures: instrument-grade suicide-risk assessment, system-level suicide prevention, empirical evaluations of chatbot crisis response, and evaluation frameworks.

Instrument-grade risk assessment. The Columbia-Suicide Severity Rating Scale (C-SSRS) [44] is the de facto standard scale for suicidal-ideation and suicidal-behaviour assessment, with good convergent and divergent validity, high sensitivity/specificity for suicidal-behaviour classifications, and sensitivity to change over time. The practical implication is that any safety layer moving beyond keyword matching — an LLM-based safety classifier, a moderation-API integration, or a structured intake — should be validated against C-SSRS rather than against an internal definition of "crisis". The keyword list is a starting point, not an endpoint.

System-level suicide prevention. [45] describe the Zero Suicide framework's seven essential elements (*Lead*, *Train*, *Identify*, *Engage*, *Treat*, *Transition*, *Improve*). The framework is a system-level analogue of what a learned scheduling policy (Section 4) would need at deployment: a chat substrate that *identifies* risk, *engages* with a crisis-specific intervention path, *transitions* to human care when indicated, and *improves* through an auditable feedback loop. A supportive (non-clinical) companion that explicitly disclaims clinical use does not need to implement the full framework, but it should be architecturally compatible with it; in our candidate implementation the SME's choices in `content/method/04-decision-rules.md` are the natural place for the *Engage*/*Transition* policy to live.

Empirical evaluations of chatbot crisis response. [46] combine field evidence of actual companion-AI interactions, a performance test of several commercial companion AIs, and a consumer-reaction experiment to risky chatbot responses; they find that (i) mental-health crises appear in a non-negligible minority of conversations; (ii) companion AIs are often *unable* to recognise or respond appropriately to signs of distress; and (iii) consumers react negatively to unhelpful or risky chatbot responses. This is the strongest published empirical justification for the two-layer safety stack rather than relying on the LLM's own judgement. [47] extend the picture with a mixed-methods comparison of licensed-therapist and general-purpose chatbot responses to scripted mental-health scenarios, and find that chatbots over-use affirming, reassuring, psychoeducational, and suggestion-style responses while under-using elaboration — and that crisis management is particularly weak (absence of risk assessment, failure to refer to hotlines).

Evaluation frameworks. [48] propose evaluation criteria for AI mental-health chatbots (adherence to ethical principles, evidence-based responses, conversational skills, safety protocols, accessibility), grounded in WHO guidance. [49] enumerate six structural concerns with ChatGPT-class tools in mental healthcare: accurate identification and diagnosis; limited understanding and misinterpretation; safety and privacy; bias and equity; lack of monitoring and regulation; and gaps in evidence and training. These two sources are the policy-and-review anchor for the ethics statement (Section 17) and for a future deployment-readiness checklist for any integrated companion.

Cross-cutting summary. The literature supports three design conclusions for any chat-based mental-health companion: (i) do not rely on the LLM's own judgement for crisis detection; (ii) explicitly disclose the AI boundary and the non-clinical scope in every conversation; (iii) maintain a human-referral path that is more visible and more frequent than any other intervention in the catalogue. The residual gap (Section 12) is that none of the published chatbot deployments combines these defaults with an *adaptive* (Decide-driven) policy over intervention selection and timing — a gap that raises both an opportunity and a new class of safety question (a learned policy that routes around crisis signals would be strictly unsafe), which any future MRT-style evaluation of an integrated companion must track from day one.

9 Cross-Cutting: Engagement, Adherence, Attrition

Engagement is the hard problem of digital mental health. The best synthesis is the [50] meta-analysis of 70 RCTs of smartphone-delivered mental-health interventions: mean short-term attrition is 24.1% (95% CI 19.3–29.6), rising to 35.5% (95% CI 26.7–45.3) at longer-term follow-up. Attrition was *significantly lower* in trials that delivered an acceptance-based intervention, offered monetary compensation, and — most directly relevant to the Decide pillar — *reminded participants to engage with the intervention.* Intervention usage typically declined over the course of the trial; no single baseline participant characteristic reliably predicted attrition. These findings are the empirical anchor for the claim we make throughout this review: engagement is the gating factor, scheduling intelligence is one of the few levers that moves it, and the Decide layer argued for in Section 4 is on the critical path to clinical gains rather than a cosmetic enhancement.

A concrete mechanism that has demonstrably moved the needle is gamification. [51] report a 3-arm RCT (eQuoo gamified app vs. non-gamified CBT journal vs. no-intervention waitlist): the gamified arm retained 21% more participants than the control arms while also producing significantly greater resilience gains. Gamification is not a substitute for a learned scheduling policy, but it is an engagement-lever a Decide policy could explicitly incorporate as one of its intervention options.

Product-design precedent: creative, non-random outreach. The design space here is not only academic. A useful mainstream product exemplar is Apple's native *Journal* app,[8] which demonstrates that proactive outreach can feel reflective rather than interruptive when it is grounded in personal context and structured for habit formation. Journal uses on-device intelligence to surface personalized writing suggestions from recent activity (e.g., photos, music, workouts, places), supports multimedia entries, and pairs scheduled notifications with streaks and insight views; it also lets users log their state of mind and save journaling time as mindful minutes in Health.[9] We cite Journal here as a *product-design precedent*, not as scientific evidence. Its relevance is that it shows a large-scale consumer product already operationalises three ideas central to the

[8] `https://www.apple.com/newsroom/2023/12/apple-launches-journal-app-a-new-app-for-reflecting-on-everyday-moments/`
[9] `https://support.apple.com/guide/iphone/journal-for-your-wellbeing-iph7b79617d5/ios`

present research agenda: context-aware prompting, multimodal reflective capture, and non-random scheduling that is meant to feel supportive rather than coercive.

10 Cross-Cutting: Ethics, Regulation, Equity, Privacy

A system that scheduled check-ins based on a learned psychological model of the client would inhabit several of the most contested axes of the digital-mental-health ethics literature. [52] frame the global-equity stake directly: LLM-powered services have the capacity to expand mental-health access at global scale, but accountability, training-data biases, and the absence of clinical oversight are substantive concerns. The Delphi consensus study of [53] crystallises the practitioner-facing ethical agenda for digital phenotyping into five priorities — privacy, transparency, consent, accountability, and fairness — with specific attention to the inadequacy of current regulatory frameworks for protecting sensitive mental-health data. [54] document how the pandemic-era relaxation of privacy and safety regulations accelerated adoption but left a structural governance gap that has persisted into the 2025–2026 deployment landscape.

For an integrated chat-based companion the implications map concretely onto common architectural choices. (i) **Consent and disclosure**: an AI-identity disclaimer (as in the candidate implementation's `content/safety/disclaimers.md`) honours the transparency principle; a future scheduling layer would need an analogous disclosure of *what is being learned about the user and how it drives the schedule.* (ii) **Privacy and data protection**: a Phase-0 no-persistence policy (e.g. `content/safety/boundaries.md`: "I do not record or remember anything between sessions" [...]) is the strongest privacy posture available and should be retained until a full privacy review accompanies any change. (iii) **Equity and access**: an opt-in, open-source, SME-configurable architecture aligns with the global-access framing of [52]; however, evaluation (below) should explicitly stratify on digital literacy and language to detect equity gaps before they widen.

A deeper ethical question that the scheduling literature itself has not yet fully faced is this: framing mental-state changes as a policy-optimisation problem (Section 4) imports the reward-function problem from reinforcement learning into a domain where the reward is not clinically uncontroversial. What counts as "success" in a given check-in depends on the SME's framework, the user's goal state, and the therapeutic contract; a policy that silently optimised for engagement at the cost of those would be strictly worse than a fixed schedule. Section 14 returns to this point.

11 Cross-Cutting: Evaluation Methodology

Evaluation of a prototype-scale integrated companion requires three instruments in combination: a proximal-effect design (the MRT), an alliance/process measure, and an engagement measure.

The MRT is already covered in Section 4: [6] introduces the design, [27] provides sample size, [28] provides standardized effect sizes, and [29] provides the data-collection checklist. The MRT is the right primary design for an adaptive-companion prototype because the scientific question is *whether and when* each push intervention component produces a proximal effect, not whether the system as a whole outperforms a waitlist control.

The alliance/process measure is the Working Alliance Inventory-Short Revised (WAI-SR) of [55]: 12 items spanning goal alignment, task agreement, and therapeutic bond. WAI-SR is standard in the chatbot-alliance literature ([34]–[36]); its adoption for an integrated chat-based companion is the default instrument for evaluating the quality of the digitally-mediated relationship. The [36] synthesis of DTA around goal/task/bond/engagement maps directly onto both WAI-SR subscales and the action space of an SME-authored Decide policy.

The engagement measure is the longitudinal retention curve, with the [50] meta-analytic baseline as the benchmark: 24.1% short-term attrition, 35.5% long-term. The proximal claim for an adaptive scheduler is that retention improves materially over fixed-cadence control while preserving (or improving) proximal-effect estimates. This is a three-way reporting matrix (retention; WAI-SR; MRT proximal effects) that a future MRT of an integrated companion should preregister.

Summary. None of these instruments are novel; all are peer-reviewed, with substantial prior use in digital-mental-health trials. The evaluation gap is not an instrument gap; it is simply that no published system

combines them in the integrated configuration the next two sections describe — which is precisely the integration claim of Section 12.

12 Synthesis: The Integration Gap

Individual progress across the three pillars and six cross-cutting concerns is substantial. *Sense* has a mature chain from ecological momentary assessment [15], [16], through passive sensing and digital phenotyping [9], [17]–[20], to text-based inference [21], [22] and LLM-based state classification [23], [24], with a standing critical caveat [25]. *Decide* has the strongest published methodological stack: a conceptual framework [5], an evaluation design [6], [27]–[29], an online-RL deployment in a real trial [7], [30], offline CMAB precedent in a mental-health-adjacent context [31], a first RCT comparing an RL-driven adaptive schedule against random and weekly baselines in a depression-relevant population [8], and a recent systematic review confirming the structural gap in JITAI deployment for mental-health outcomes specifically [26]. *Act* has three peer-reviewed RCTs of conversational-agent interventions [1]–[3], a 66-study meta-analytic baseline [4], and a mature digital-therapeutic-alliance literature [34]–[36].

Integration criteria. Within the corpus surveyed here, we found no peer-reviewed system that satisfies *all* of the following five operational criteria simultaneously:

- (C1) A *learned, longitudinal* client model derived from conversation and/or light-touch EMA — not a static intake form or a single self-report.
- (C2) A *theory-grounded scheduling policy* that jointly selects *when* to check in and *what* content to offer, using JITAI / contextual-bandit / RL math [5]–[7].
- (C3) A *multi-framework intervention library* the SME authors and the policy selects from, aligned to the BCT taxonomy [33] and rooted in CBT, ACT, MI, or related evidence bases [38]–[40].
- (C4) Conversational delivery with a *crisis-safe fallback* and *explicit AI-disclosure*, informed by C-SSRS [44] and Zero Suicide [45], with documented empirical awareness of LLM-safety gaps in mental-health conversations [46]–[48].
- (C5) An *open / inspectable* architecture evaluable on listener-centred, MRT-style outcomes (proximal effects; WAI-SR alliance [55]; engagement/retention curves [50]).

What the closest-related systems cover. Closely related work covers strict subsets. HeartSteps and its RL extension [7], [30] satisfy C1+C2 brilliantly but for physical activity, not for mental-health content, and do not use a chat substrate. Woebot and Wysa [1], [2] satisfy C3+C4 at small scale but are rule-based (no C1/C2 in any meaningful sense) and are single-framework. Therabot's NEJM AI RCT [3] is the strongest single data point for C3+C4 with Gen-AI, but uses a fixed delivery cadence rather than a learned schedule; the DIAMANTE RCT of [8] occupies the other corner (C1+C2 in a depression-relevant population) but is message-based, not conversational. TalkPlay-style LLM-orchestrated systems satisfy C2 partially but do not learn a longitudinal client model. The [26] systematic review makes the same point at the field level: JITAI deployment in mental health specifically remains thin despite an otherwise mature methodological base.

Table 3 lists one prior-system baseline per axis against which a future integrated-companion prototype would be benchmarked.

Our proposed solution. We have been referring throughout the pillar sections to a candidate implementation. We now name it: **Willow**, an open-source conversational mental-health companion released at `https://github.com/memari-majid/willow` and deployed at `https://willowspace.dev`. Willow's distinctive architectural commitment is *layering C1+C2 on top of an existing C3+C4 stack while holding C5 as a hard design constraint*. The SME-authored `content/` directory already provides the multi-framework catalogue (C3) [38]–[40]; the `src/lib/ai/` layer provides conversational delivery (C4) with explicit AI-disclosure; crisis safety (C4) is anchored to a baseline keyword detector and a system-prompt reminder chain. What is missing in the current scaffold is the Sense→Decide loop, and the next section argues it is feasible to add without new model training.

Implementation trend and workflow vocabulary. From an engineering perspective, the proposed system is best understood as an *AI agent* or *agentic workflow*, not merely as a single chatbot prompt. In

Axis	Closest baseline / what we test against it
Sense (C1)	StudentLife [19], [20]; Mental-LLM [23]. Time-to-stable-client-model; calibration and interpretability at the individual level.
Decide (C2)	Personalized HeartSteps [7]; DIAMANTE RL [8]; CMAB-for-emotion-regulation [31]. Proximal-effect gain over random-cadence and fixed-cadence baselines, plus retention gain.
Act (C3/C4)	Therabot [3]; Woebot [1]; Wysa [2]. Symptom-scale change at 4–8 weeks; WAI-SR digital therapeutic alliance [55].
Crisis (C4)	Keyword + C-SSRS [44] + LLM moderation against [46], [47]. False-negative rate on scripted escalating-risk scenarios.
Engagement	[50] meta-analytic baseline (24% short-term, 36% long-term attrition). Retention-curve improvement of an adaptive schedule over fixed-cadence control.

Table 3: Comparison shortlist for a future integrated-companion prototype evaluation.

Subsystem	Existing evidence / primary risk
Sense	LLM-based state inference on chat data [23], [24]; interpretable lexical features [21], [22]; optional EMA / wearable layer [9], [10], [15], [16]. *Risk:* hallucinated specificity in the language layer; calibration failure at individual scale; false positives from noisy wearable signals; ethical concerns about state-inference as surveillance [25].
Decide	JITAI framework [5]; MRT [6], [27]; online RL [7]; offline CMAB in emotion regulation [31]; mental-health RCT precedent [8]; behaviour-change taxonomies [32], [33]. *Risk:* reward-specification error (optimising engagement at the expense of clinical value); policy instability at small N.
Act	Already shipped in `src/lib/ai/` with the `content/` Markdown system-prompt assembly; RCT precedent across three agents [1]–[3]; meta-analytic support for CBT-based chat [4]. *Risk:* alliance drift [34], [36]; process-vs-content confusion.
Crisis	Keyword detector (already shipped in `content/safety/`); moderation API; LLM safety classifier; validation against C-SSRS [44]. *Risk:* false negatives (catastrophic); keyword list staleness.
Evaluation	MRT [6], [27]–[29]; WAI-SR [55]; attrition-curve benchmark [50]. *Risk:* human evaluation cost; longitudinal follow-up attrition; small N.
Ethics / governance	WHO-aligned ethical principles [53], [54]; transparent consent; SME-owned content; open-weights architecture [52]. *Risk:* regulatory drift between FDA "wellness app" and "SaMD" classifications as capability grows.

Table 4: Feasibility assessment of a candidate integrated system satisfying criteria C1–C5.

current practitioner vocabulary, this means an automated workflow that combines triggers, branching logic, memory, tool invocation, and human-in-the-loop escalation. Workflow-orchestration tools such as *n8n*[10] and *LangGraph*[11] have popularized exactly this graph-based implementation style. We do *not* cite these tools as scientific evidence for efficacy; rather, they are relevant implementation precedents showing how the proposed Sense→Decide→Act loop can be operationalized as an automated, inspectable, agentic workflow. Figure 1 illustrates this implementation view.

13 Feasibility of a Candidate Integrated System

The question is whether every subsystem in a C1–C5-compliant integrated system has a credible literature-backed implementation path that avoids new model training. Table 4 summarises.

The key result is straightforward: **none of the subsystems requires training a new foundation model.** The remaining work is an integration problem plus a human-subjects evaluation. The review also supports a specific implementation order: a *Decide-first* prototype, because the policy layer is the narrowest subsystem with the strongest low-cost evaluation design (the MRT), and because the Sense substrate for a

[10] `https://n8n.io/`
[11] `https://langchain-ai.github.io/langgraph/`

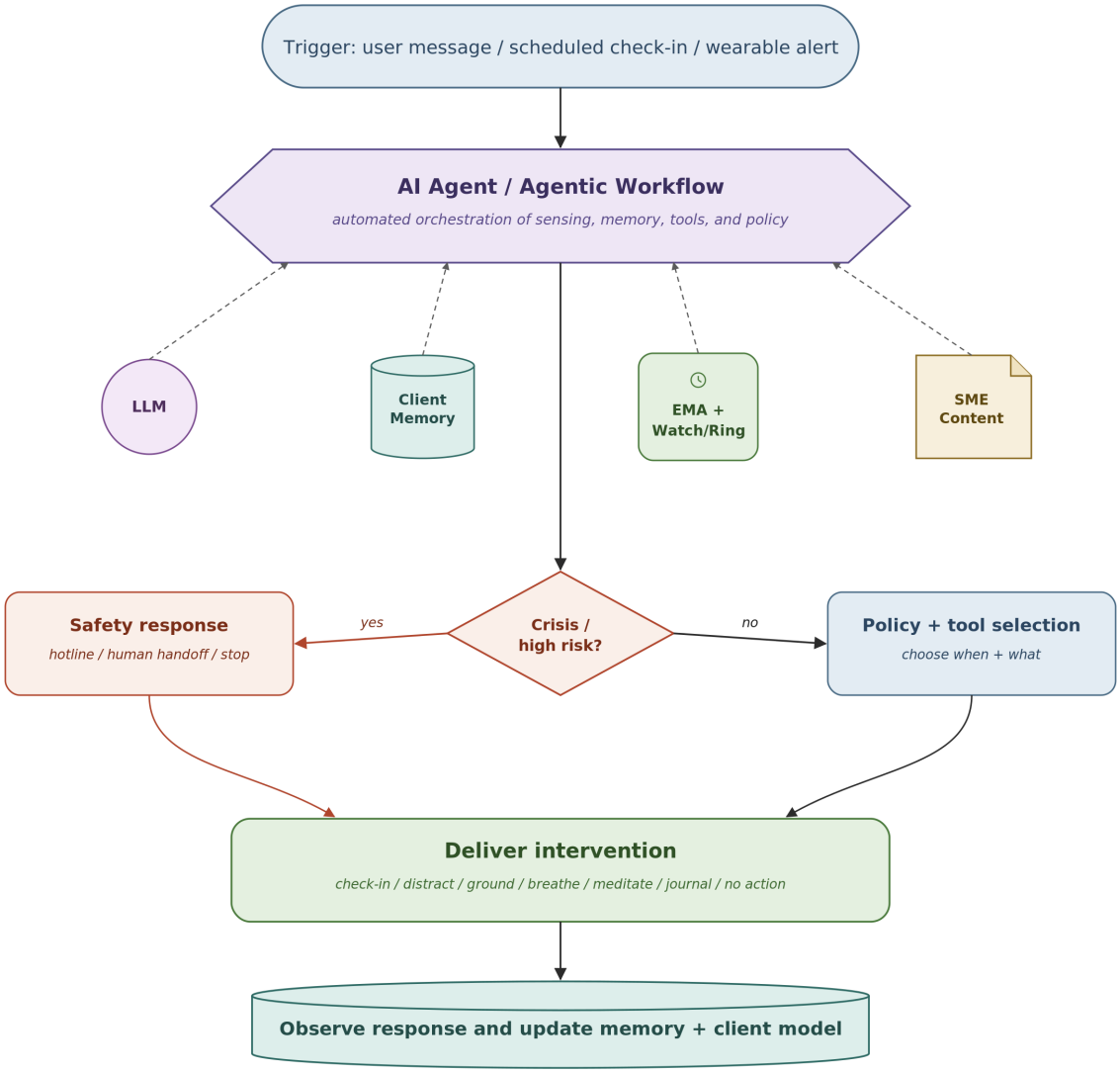

Figure 1: Proposed automated AI-agent / agentic-workflow view of the project. A trigger (user message, scheduled check-in, or wearable alert) enters an orchestration node that combines an LLM, client memory, EMA / watch–ring signals, and SME-authored content; a risk gate branches to crisis handling when needed; otherwise a policy selects and delivers a low-risk intervention, and the response updates the client model for the next loop. Wearables are a future, opt-in extension rather than the baseline substrate.

chat-only deployment is already available via the same LLM that powers *Act*.

14 Limitations

As a scoping review with feasibility framing, this paper has five explicit limitations.

(i) *Single-reviewer screening.* A single reviewer screened every record returned by the twelve Boolean queries; PRISMA-2020 and PRISMA-ScR both recommend double-screening, and a future revision should add an independent screener and report inter-rater agreement [13], [14].

(ii) *No formal risk-of-bias scoring.* The review synthesises evidence narratively across HCI, mHealth biostatistics, clinical psychology, behaviour-change science, and NLP; we did not apply a quantitative risk-of-bias instrument because no single instrument is appropriate across these heterogeneous evaluation designs.

(iii) *Industry sources as dated snapshots.* Documented chatbot-harm cases (Replika, NEDA Tessa, Character.ai) are surfaced by [46] and [47] rather than being cited as primary industry sources; we have not independently verified press reports and do not treat them as load-bearing.

(iv) *Scope.* The review covers English-language, adult, text-chat-modality digital mental health. Multilingual deployment, adolescent and older-adult populations, voice/video modalities, and clinical-grade care (therapist replacement vs. augmentation) are each their own review; they are flagged as priorities for future revisions.

(v) *Ethical asymmetry of the framing.* Treating mental-state changes as a *policy-optimisation problem* (Section 4) imports the reward-specification problem from reinforcement learning into a domain where the reward is not clinically uncontroversial. We discuss this in Section 10; a future revision should engage the clinical-ethics literature more fully than a scoping-review format allows.

15 Conclusion

This review establishes three results. *First*, within the corpus surveyed here, no peer-reviewed system jointly satisfies the five operational criteria C1–C5 of an integrated, learned, theory-grounded, scheduled check-in workflow for a conversational mental-health companion (Section 12). *Second*, the empirical literature supports the design choices such a system would commit to: a longitudinal chat-derived client model [20], [23], a JITAI-style scheduling policy with MRT evaluation [5], [6], conversational delivery with explicit crisis-safe fallbacks [3], [46], and engagement-aware reporting alongside proximal effects [50]. *Third*, every technical component required to fill the gap is already available as open weights, open source, or open methodology, making the integration tractable without any new model training (Table 4).

Status of the present work. This paper is a literature review and a research agenda. It does not report a built prototype, a user study, or any clinical or behavioural outcomes. The candidate open-source implementation (Willow, `https://github.com/memari-majid/willow`) currently realises only the *Act* layer (C3+C4) plus a Phase-0 keyword crisis stack; the *Sense* layer (C1) and the *Decide* policy (C2) that this review motivates have not yet been implemented. We make this status explicit so that the review is read as the first contribution in a multi-step programme rather than as the announcement of a completed system.

Proposed research agenda. The recommended next step is a *Decide-first prototype* followed by a micro-randomized trial:

- **Stage 1 (this paper).** Lit-review-driven scoping and research agenda. Establish C1–C5, the comparison shortlist, and the feasibility map. *Status: this paper.*
- **Stage 2 (next).** Implement the Decide layer: a small Thompson-sampling contextual-bandit scheduler over Willow's existing chat substrate that selects (a) when to proactively reach out and (b) which technique to offer from the SME-authored `content/techniques/` catalogue.
- **Stage 2b (optional sensing extension).** Add an explicit-consent wearable branch to the Sense layer, using watch or ring signals (e.g., sleep, activity, HRV, temperature, recovery proxies) as timing cues for low-risk interventions such as distraction, grounding, breathing, or meditation prompts.

- **Stage 3.** A micro-randomized trial [6], [27], [29] at $N \approx 30$–50 participants benchmarked against the comparison shortlist of Table 3, with the three-way reporting matrix of Section 11 (MRT proximal effects; WAI-SR alliance; retention curves).
- **Stage 4.** A larger pre-registered RCT, comparable in design to [8] and the NEJM AI Therabot trial of [3].

More generally, we argue that this problem belongs at the intersection of digital-mental-health biostatistics, HCI, clinical psychology, and AI safety, and that the SME↔developer content-separation architecture Willow already implements is a defensible substrate for a research agenda that takes all four of those disciplines seriously. The present paper contributes the review and the agenda; the prototype, the MRT, and the RCT are sequential next steps.

16 AI Usage Statement

We disclose the following uses of AI tools in the preparation of this manuscript. (i) *Verified literature search.* The Scite MCP (`https://api.scite.ai`) was used to retrieve DOI metadata, full-text excerpts, smart citation context (supporting / contrasting / mentioning / unclassified), and editorial-notice flags for every cited source. Every citation in this paper has a DOI independently verified against Scite's index; no citation was generated, invented, or paraphrased from an LLM's parametric memory. (ii) *Drafting and synthesis assistance.* An LLM-based coding assistant (Claude) operating inside the Cursor IDE was used to assist with prose drafting, reorganisation, and synthesis across sections of this review. All research questions, inclusion/exclusion decisions, the taxonomy, claims, comparative analyses, feasibility judgements, and conclusions were authored and verified by the human author(s), who assume full responsibility for the scientific content and reference validity of this paper. (iii) *Editing.* Standard grammar correction and clarity editing tools were used in a copy-editing role. The review reports no new experimental results; no AI tool was used to generate or alter experimental data, analyses, or figures.

17 Ethics Statement

This is a literature review and contains no new human-subjects data; Institutional Review Board approval was not required for this work, and no participants were recruited. The review nonetheless touches four ethics-relevant areas that any future system building on this synthesis must address.

(i) *Mental-state inference as a surveillance risk.* Any system that implements Pillar A (Sense) learns an inferred state representation of the user; the sociological critique of [25] and the ethical-governance recommendations of [53] together set the minimum bar: the representation must be inspectable by the user, resettable on request, and never shared or sold.

(ii) *Engagement-optimisation as a misaligned reward.* The Decide pillar treats intervention scheduling as a policy-optimisation problem (Section 4). The reward function matters more than the algorithm. A policy that silently optimised for engagement at the cost of clinical value would be strictly worse than a fixed schedule; any future prototype — Willow included — must explicitly separate *engagement proxies* from *therapeutic-goal proxies* in the reward, and report both.

(iii) *Crisis-handoff responsibility.* The candidate implementation (Willow) is explicitly *not* a therapist, doctor, or crisis service (`content/safety/disclaimers.md`); the crisis layer exists to route users to human help, not to substitute for it. The empirical literature [46], [47] shows that general-purpose chat substrates are not safe defaults for crisis, which is why the keyword detector (Phase 0) and any downstream LLM-safety classifier (Phase 4) must be held to a different evaluation standard than the rest of the action catalogue: missed signals are not acceptable failure modes.

(iv) *Scope and equity.* The review's restriction to English-language, adult, text-chat-modality interventions is a scope limitation, not a normative claim. The global-equity framing of [52] and the pandemic-era digital access shifts of [54] both argue that access is itself a therapeutic outcome; an evaluation of the proposed prototype must stratify on digital literacy and language to avoid widening the very gap it aims to close.

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